

Projecting Softness: Finger-Bending Illusions in Tangible Projection Mapping

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Abstract

We present a projection-based technique that visually induces the illusion of finger bending to enhance the perception of softness when touching rigid physical objects. Our method uses luminance-based motion differences to create the impression that the finger is naturally bending in response to contact, even though no physical deformation occurs. Evaluation results from subjects handling 3D objects showed that our method significantly enhances the perception of softness compared to existing techniques. This approach requires no wearable devices and broadens the possibilities for haptic-free material expression in augmented reality.

CCS Concepts

• **Human-centered computing** → **Virtual reality; Mixed / augmented reality.**

Keywords

Softness Illusion, Finger-Bending Illusion, Dynamic Projection Mapping, Tangible Augmented Reality, Crossmodal Perception

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1 Introduction

Dynamic projection mapping enables realistic visual augmentation of physical objects by projecting real-time graphics onto their surfaces. While previous works have explored material appearance manipulation, conveying tactile softness using projection remains a challenge—particularly when the object is rigid and physically grasped [Halvorson et al. 2022]. In such cases, visual softness conflicts with tactile hardness, breaking the illusion. While color-based cues such as those used in SoftAR [Punpongpanon et al. 2015] can amplify the perceived softness, they assume that a soft object is present. In contrast, our setting involves completely rigid objects, where such cues are less effective or inapplicable.

2 Proposed method

We propose a projection-based technique that creates a convincing visual illusion of softness by combining two synchronized effects: finger-bending and surface indentation (Figure 1). These effects are tightly coupled to simulate the visual response of both the user's finger and the object's surface during contact. By projecting luminance-based motion differences onto the finger and rendering localized surface deformation, the technique induces a crossmodal illusion of pressing one's finger into a soft material, even though the object is physically rigid.

The finger-bending illusion builds upon luminance difference projection techniques [Fujimoto et al. 2020], which visually simulate deformation through subtle pixel-level motion. In our method, we adapt this principle to dynamic hand-object interactions, projecting the motion difference between a real hand and a flexed hand model onto the user's actual finger.

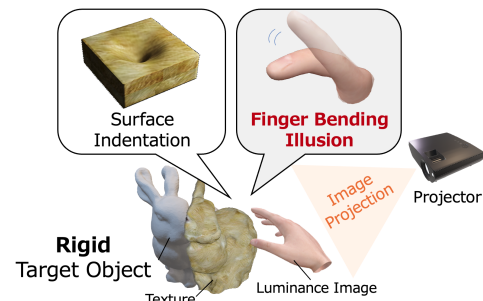


Figure 1: Overview of our method.

2.1 Finger Bending Illusion

To achieve the finger-bending illusion, we first capture a live image of the user's hand using an infrared camera to avoid interference from the projector (Figure 2(a)). Simultaneously, a 3D hand model matched to the user's hand is rendered with a slightly flexed posture, simulating one or more fingers pressing into a soft object (Figure 2(b)). The difference between the real-time image and the simulated flexed image is calculated pixel-wise (Figure 2(c)). This differential image, containing subtle luminance-based motion cues, is projected directly onto the user's fingers.

This process causes the user to perceive a flexure motion as if their finger is bending or sinking into the object, despite the hand actually remaining in a more neutral grasping posture. By carefully calibrating the lighting and synchronizing the projection timing, we create a compelling illusion that exploits the brain's visual dominance in perceiving body posture.

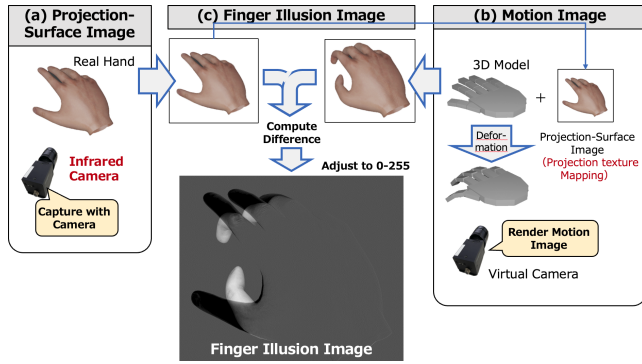


Figure 2: Process for generating a finger illusion image.

2.2 Surface Deformation Rendering

To complement the finger illusion, we render a visually plausible indentation on the object's surface at the contact point. Using a physics engine (Bullet Physics), we simulate how a soft object would deform under finger pressure. This is done by estimating the finger tip's position and applying a capsule-shaped collider that mimics finger geometry.

The resulting deformation is textured and shaded under a virtual light source aligned with the projector's position to reinforce depth cues. This surface deformation is projected onto the physical object in real time and synchronized with the finger-bending illusion, creating a unified impression of interaction with a soft material.

By combining these two illusions—finger bending and surface indentation—we amplify the softness perception without the need for haptic actuators (Figure 3).

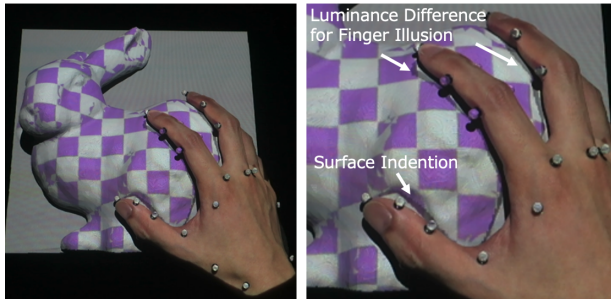


Figure 3: Projection of two synchronized illusion images.

3 Results

We recruited 11 subjects to evaluate four presentation conditions—indentation only, a prior method [Punpongson et al. 2015], our combined illusion method, and a hybrid of both. Each condition was tested across six combinations of two rigid object shapes (a hemispherical object and the Stanford Bunny, with hard and soft visual impressions) and three surface textures (checker pattern, short fur, and long fur) (Figure 4). In the prior method, finger regions were colored red in proportion to the simulated indentation force; in the hybrid method, red coloring was modulated by the finger

bending amount produced by our technique. Participants rated the perceived softness relative to a baseline condition showing only the finger-bending illusion (set as 1.0).

Figure 5 shows the results of the user testing. While the prior method did not significantly enhance the perception of softness compared to the indentation-only condition, our combined illusion method significantly increased the perception of softness in many cases. Furthermore, combining the prior method with our method did not result in a notable synergistic effect. Future studies should explore a wider variety of object geometries and textures to further validate the generalizability of this approach.

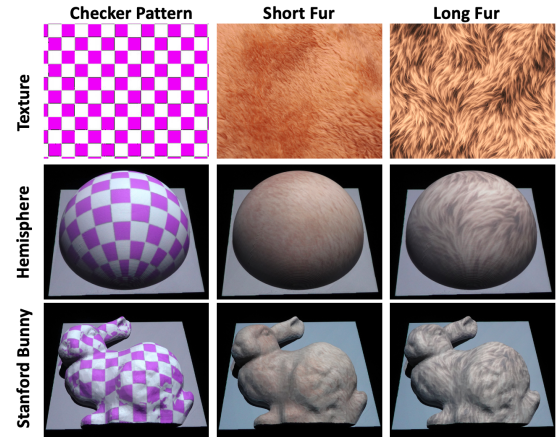
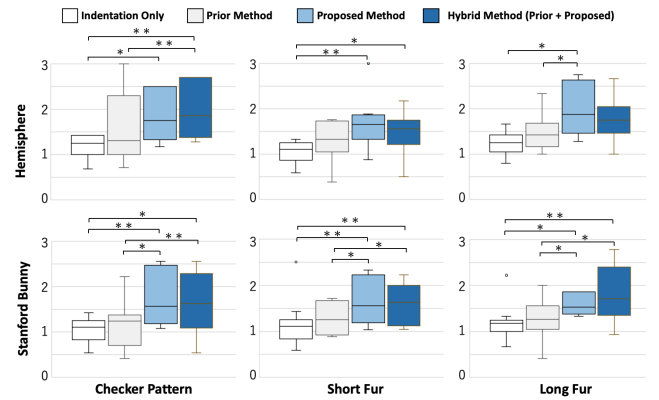


Figure 4: Grasped object and texture combinations.

Figure 5: Results for softness perceived by test subjects. * $p < 0.05$, ** $p < 0.01$

References

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